THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME — SET 1

Chapter 6: **Triangles** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 6	BPT
2	**(a)** 9	Sides in ratio $\sqrt{\text{areas}}$
3	**(a)** 18	$d^2 = s^2 + s^2$
4	**(a)** 25	Pythagoras
5	**(d)** right-angled	Check $9^2 + 12^2$ against 15^2
6	**(c)** 5	$\frac{AE}{AC} = \frac{AD}{AB}$
7	**(b)** 12	BPT
8	**(b)** 12	Sides in ratio $\sqrt{\text{areas}}$
9	**(a)** 50	$d^2 = s^2 + s^2$
10	**(c)** 40	Pythagoras

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$d^2 = 289 - 225 = 64$ (1); $d = 8$ m (1).	1+1
12	$25^2 = 625$; sum of squares of others $= 625$ (1); equal, so right-angled (1).	1+1
13	Side ratio $= 1:2$ (1); area ratio $= 1:4$ (1).	1+1
14	$d^2 = 625 - 576 = 49$ (1); $d = 7$ m (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	By the Basic Proportionality Theorem, $\frac{AD}{DB} = \frac{AE}{EC}$ (1); $\frac{4}{6} = \frac{12}{EC}$ (1); $EC = 18$ cm (1).	1+1+1
16	Ratio of areas $=$ (ratio of sides)^2 (1); $\frac{16}{36} = \frac{4^2}{x^2}$ (1); $x = 6$ cm (1).	1+1+1
17	$\triangle ADE \sim \triangle ABC$, $AD:AB = 1:3$ (1); area ratio $= 1:9$ (2); trapezium area $= 8$ parts, so $\triangle ADE : DBCE = 1:8$ (2).	1+2+2
	OR: By the Basic Proportionality Theorem, $\frac{AD}{DB} = \frac{AE}{EC}$ (1); $\frac{5}{3} = \frac{10}{EC}$ (1); $EC = 6$ cm (1).	1+1+1
18	Ratio of areas $=$ (ratio of sides)^2 (1); $\frac{18}{162} = \frac{3^2}{x^2}$ (1); $x = 9$ cm (1).	1+1+1
	OR: $\triangle ADE \sim \triangle ABC$, $AD:AB = 2:5$ (1); area ratio $= 4:25$ (2); trapezium area $= 21$ parts, so $\triangle ADE : DBCE = 4:21$ (2).	1+2+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Given $\triangle ABC$, $DE \parallel BC$; join BE, CD and draw $EM \perp AB$, $DN \perp AC$ (1); $\frac{ar(ADE)}{ar(BDE)} = \frac{\frac{1}{2}AD \cdot EM}{\frac{1}{2}DB \cdot EM} = \frac{AD}{DB}$ and $\frac{ar(ADE)}{ar(CDE)} = \frac{AE}{EC}$ (2); $ar(BDE) = ar(CDE)$ (same base DE, same parallels) (1); hence $\frac{AD}{DB} = \frac{AE}{EC}$ (1).	1+2+1+1
	OR: Ratio of sides $=$ ratio of perimeters $= 1:2$ (1); side $= 10$ cm (1); ratio of areas $= 1:4$ (1).	1+1+1
20	If $\frac{AD}{DB} = \frac{AE}{EC}$, then $DE \parallel BC$ (1). Suppose not; draw $DE' \parallel BC$ meeting AC at E' (1); by BPT $\frac{AD}{DB} = \frac{AE'}{E'C}$ (1); so $\frac{AE}{EC} = \frac{AE'}{E'C}$, hence $E = E'$ (1); thus $DE \parallel BC$ (1).	1+1+1+1+1
	OR: Ratio of sides $=$ ratio of perimeters $= 1:3$ (1); side $= 18$ cm (1); ratio of areas $= 1:9$ (1).	1+1+1

Mistakes to avoid:

- Writing the BPT as $\frac{AD}{AB} = \frac{AE}{EC}$; mix of a part-to-whole and a part-to-part ratio.
- Equating the *areas* ratio to the *sides* ratio instead of its square.
- Matching vertices in the wrong order while writing similarity ($\triangle ABC \sim \triangle PQR$ must have A to P, B to Q, C to R).
- Applying the Pythagoras theorem with the wrong side as hypotenuse (hypotenuse is opposite the right angle and is the longest side).

- Using the converse of Pythagoras without squaring all three sides, or comparing with the wrong side.
- Saying two triangles are similar because they have equal ratio of only two pairs of sides (SAS needs the included angle).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 6: **Triangles** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** not right-angled	Check $7^2 + 8^2$ against 10^2
2	**(d)** 8	$\frac{AE}{AC} = \frac{AD}{AB}$
3	**(b)** 12	BPT
4	**(c)** 6	Sides in ratio $\sqrt{\text{areas}}$
5	**(d)** 72	$d^2 = s^2 + s^2$
6	**(a)** 15	Pythagoras
7	**(d)** right-angled	Check $8^2 + 15^2$ against 17^2
8	**(d)** 5	$\frac{AE}{AC} = \frac{AD}{AB}$
9	**(a)** 36	BPT
10	**(a)** 15	Sides in ratio $\sqrt{\text{areas}}$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$10^2 = 100$; sum of squares of others $= 85$ (1); not equal, so not right-angled (1).	1+1
12	Side ratio $= 1:3$ (1); area ratio $= 1:9$ (1).	1+1
13	$d^2 = 841 - 441 = 400$ (1); $d = 20$ m (1).	1+1
14	$85^2 = 7225$; sum of squares of others $= 7225$ (1); equal, so right-angled (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	By the Basic Proportionality Theorem, $\frac{AD}{DB} = \frac{AE}{EC}$ (1); $\frac{2}{7} = \frac{10}{EC}$ (1); $EC = 35$ cm (1).	1+1+1
16	Ratio of areas $=$ (ratio of sides)^2 (1); $\frac{25}{64} = \frac{5^2}{x^2}$ (1); $x = 8$ cm (1).	1+1+1
17	$\triangle ADE \sim \triangle ABC$, $AD:AB = 1:4$ (1); area ratio $= 1:16$ (2); trapezium area $= 15$ parts, so $\triangle ADE : DBCE = 1:15$ (2).	1+2+2
	OR: By the Basic Proportionality Theorem, $\frac{AD}{DB} = \frac{AE}{EC}$ (1); $\frac{6}{4} = \frac{12}{EC}$ (1); $EC = 8$ cm (1).	1+1+1
18	Ratio of areas $=$ (ratio of sides)^2 (1); $\frac{24}{\frac{200}{3}} = \frac{6^2}{x^2}$ (1); $x = 10$ cm (1).	1+1+1
	OR: $\triangle ADE \sim \triangle ABC$, $AD:AB = 3:7$ (1); area ratio $= 9:49$ (2); trapezium area $= 40$ parts, so $\triangle ADE : DBCE = 9:40$ (2).	1+2+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Let $\triangle ABC \sim \triangle PQR$; draw altitudes AM and PN (1); $\frac{ar(ABC)}{ar(PQR)} = \frac{BC \cdot AM}{QR \cdot PN}$ (1); $\triangle ABM \sim \triangle PQN$ gives $\frac{AM}{PN} = \frac{AB}{PQ}$ (1); with $\frac{AB}{PQ} = \frac{BC}{QR}$ get $\left(\frac{AB}{PQ}\right)^2$, and equal to $\left(\frac{BC}{QR}\right)^2 = \left(\frac{CA}{RP}\right)^2$ (2).	1+1+1+2
	OR: Ratio of sides $=$ ratio of perimeters $= 1:4$ (1); side $= 16$ cm (1); ratio of areas $= 1:16$ (1).	1+1+1
20	Let $\angle B = 90°$; draw $BD \perp AC$ (1); $\triangle ADB \sim \triangle ABC$ gives $AB^2 = AD \cdot AC$ (1); $\triangle BDC \sim \triangle ABC$ gives $BC^2 = CD \cdot AC$ (1); adding, $AB^2 + BC^2 = AC(AD + DC) = AC^2$ (2).	1+1+1+2
	OR: Ratio of sides $=$ ratio of perimeters $= 1:5$ (1); side $= 15$ cm (1); ratio of areas $= 1:25$ (1).	1+1+1

Mistakes to avoid:

- Writing the BPT as $\frac{AD}{AB} = \frac{AE}{EC}$; mix of a part-to-whole and a part-to-part ratio.
- Equating the *areas* ratio to the *sides* ratio instead of its square.
- Matching vertices in the wrong order while writing similarity ($\triangle ABC \sim \triangle PQR$ must have A to P, B to Q, C to R).
- Applying the Pythagoras theorem with the wrong side as hypotenuse (hypotenuse is opposite the right angle and is the longest side).

- Using the converse of Pythagoras without squaring all three sides, or comparing with the wrong side.
- Saying two triangles are similar because they have equal ratio of only two pairs of sides (SAS needs the included angle).

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 6: **Triangles** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 128	$d^2 = s^2 + s^2$
2	**(d)** 50	Pythagoras
3	**(d)** not right-angled	Check $10^2 + 11^2$ against 15^2
4	**(b)** 7	$\frac{AE}{AC} = \frac{AD}{AB}$
5	**(d)** 10	BPT
6	**(a)** 15	Sides in ratio $\sqrt{\text{areas}}$
7	**(b)** 162	$d^2 = s^2 + s^2$
8	**(b)** 35	Pythagoras
9	**(c)** right-angled	Check $20^2 + 48^2$ against 52^2
10	**(b)** 7	$\frac{AE}{AC} = \frac{AD}{AB}$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Side ratio $= 1:4$ (1); area ratio $= 1:16$ (1).	1+1
12	$d^2 = 1681 - 1600 = 81$ (1); $d = 9$ m (1).	1+1
13	$14^2 = 196$; sum of squares of others $= 181$ (1); not equal, so not right-angled (1).	1+1
14	Side ratio $= 1:5$ (1); area ratio $= 1:25$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	By the Basic Proportionality Theorem, $\frac{AD}{DB} = \frac{AE}{EC}$ (1); $\frac{3}{8} = \frac{9}{EC}$ (1); $EC = 24$ cm (1).	1+1+1
16	Ratio of areas $=$ (ratio of sides)^2 (1); $\frac{12}{147} = \frac{2^2}{x^2}$ (1); $x = 7$ cm (1).	1+1+1
17	$\triangle ADE \sim \triangle ABC$, $AD:AB = 2:7$ (1); area ratio $= 4:49$ (2); trapezium area $= 45$ parts, so $\triangle ADE : DBCE = 4:45$ (2).	1+2+2
	OR: By the Basic Proportionality Theorem, $\frac{AD}{DB} = \frac{AE}{EC}$ (1); $\frac{7}{2} = \frac{14}{EC}$ (1); $EC = 4$ cm (1).	1+1+1
18	Ratio of areas $=$ (ratio of sides)^2 (1); $\frac{54}{96} = \frac{9^2}{x^2}$ (1); $x = 12$ cm (1).	1+1+1
	OR: $\triangle ADE \sim \triangle ABC$, $AD:AB = 3:5$ (1); area ratio $= 9:25$ (2); trapezium area $= 16$ parts, so $\triangle ADE : DBCE = 9:16$ (2).	1+2+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Given $AC^2 = AB^2 + BC^2$ (1); construct $\triangle PQR$ with $\angle Q = 90^\circ$, $PQ = AB$, $QR = BC$ (1); $PR^2 = PQ^2 + QR^2 = AB^2 + BC^2 = AC^2$, so $PR = AC$ (1); $\triangle ABC \cong \triangle PQR$ (SSS), hence $\angle B = \angle Q = 90^\circ$ (2).	1+1+1+2
	OR: Ratio of sides $=$ ratio of perimeters $= 1:2$ (1); side $= 16$ cm (1); ratio of areas $= 1:4$ (1).	1+1+1
20	D, E midpoints of AB, AC (1); $\frac{AD}{DB} = 1 = \frac{AE}{EC}$ (2); by the converse of BPT, $DE \parallel BC$ (2).	1+2+2
	OR: Ratio of sides $=$ ratio of perimeters $= 1:3$ (1); side $= 21$ cm (1); ratio of areas $= 1:9$ (1).	1+1+1

Mistakes to avoid:

- Writing the BPT as $\frac{AD}{AB} = \frac{AE}{EC}$; mix of a part-to-whole and a part-to-part ratio.
- Equating the *areas* ratio to the *sides* ratio instead of its square.
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